BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 **Data Analysis Report**

Project ID: 25-26J-148

Project Title: Player Based Segmentation and Wicket Type Detection In Cricket Videos

1. Introduction

1.1. Background:

The rapid growth of digital media platforms has led to a significant increase in the availability of full-length sports videos, particularly cricket match recordings. Cricket is one of the most widely followed sports, generating extensive video content used by fans, coaches, analysts, broadcasters, and training institutions. However, these videos are typically long, unstructured, and lack effective mechanisms for player-based navigation and event-specific analysis. [2]

Traditional methods of reviewing cricket match videos rely heavily on manual searching and visual inspection to identify important moments such as fours, sixes, wickets, and player-specific performances. This manual approach is time-consuming, inefficient, and impractical for large-scale video analysis. Additionally, tracking batsman and bowler changes, identifying key match events, and generating player-wise highlights require significant human effort and technical expertise.

Recent advances in data science, computer vision, and machine learning have enabled automated analysis of complex visual data. Broadcast cricket videos contain rich visual cues such as scoreboard information, umpire gestures, and player movements, which can be leveraged to extract meaningful insights. [1] Techniques such as Optical Character Recognition (OCR) can be used to extract player names and match context from scoreboards, while deep learning models can detect umpires and classify event-specific gestures. [3][4]

This research focuses on developing an intelligent cricket video analysis system that segments match videos based on player identity and key events. By combining OCR, computer vision, and machine learning techniques, the proposed approach aims to transform unstructured cricket videos into structured, searchable content. The system enables efficient player-based navigation, accurate event detection, and automated highlight generation, contributing to improved sports analytics, content management, and match review processes. [5]

1.2. Research Problem:

The research problem addresses the increasing need for efficient analysis and management of full-length cricket match videos, driven by the demand of fans, coaches, analysts, and broadcasters for quick access to player-specific moments and key events. Traditional methods of reviewing matches rely heavily on manually scrolling through videos to locate important events such as fours, sixes, wickets, and player performances. This manual process is time-consuming, labor-intensive, and often leads to missed critical moments, making effective video analysis inefficient.

Manual approaches require extensive human effort to identify batsmen, bowlers, and events, track player changes during overs, link events with timestamps, and generate highlights. Furthermore, the lack of structured video data prevents systematic player-wise analysis and rapid content retrieval. As cricket matches involve dynamic and fast-changing scenarios, it is impractical to rely solely on manual methods for accurate and timely event detection.

As a sustainable solution, automated cricket video analysis using data-driven techniques such as computer vision, optical character recognition (OCR), and deep learning can efficiently extract player information, detect key events, and link players to moments with precise timestamps. Broadcast videos provide rich visual cues including scoreboard information, umpire gestures, and player movements, which can be leveraged to develop a scalable and reliable system for player-based video segmentation.

By automating event detection and player-based indexing, the proposed approach aims to reduce manual effort, improve navigation efficiency, enable rapid generation of highlights, and provide actionable insights for fans, coaches, analysts, and content creators.

1.3. Objectives:

IT Number	Objective	Objective number
IT22141606	Segment cricket videos based on player identity	1

2. Data Exploration

2.1. Data Collection

The dataset for this research was collected from publicly available broadcast cricket match videos sourced primarily from YouTube. Full-length cricket match recordings were downloaded to ensure diverse match conditions, including different teams, venues, and camera angles.Match videos were then divided into individual frames to enable precise temporal analysis of player actions and match events.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
Data source 1 • Player identification dataset https://docs.google.com/spreadsheets/d/1QNcbmWh52foKajmPC9eoI_1K_svmPM8H/edit?usp=sharing&ouid=106992203468750154626&rtpof=true&sd=true	OCR extracted scoreboard text with manually labeled batsman and bowler names	Broadcast cricket match videos (YouTube)	3,800 records	Extracted text, Batsman 1, Batsman 2, Bowler, Timestamp
• Umpire,non umpire dataset https://drive.google.com/drive/folders/1b3XtnBcuF79UdKgtAJK	Manually captured screenshots to identify umpire presence	Broadcast cricket match videos (YouTube)	2,000 images	Image, Label (Umpire / Non-Umpire)

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• Umpire event detection dataset https://drive.google.com/drive/folders/1gwGDlmzxPmeFIAoyKVefNBeI7MhcTn3O?usp=sharing	Manually captured umpire images labeled by event type	Broadcast cricket match videos (YouTube)	2,000 images	Image, Event label (4 / 6 / Wicket / No Action)

2.3. Suitability Analysis

2.3.1.Relevance to Individual Research Objectives:

	1	2	3	4
Data source 1	x			

3. Methodology
   3.1. Data Preprocessing:

Objective 1 - IT22141606

	Transformation technique						
	Data Cleaning	Data Normalization	Data encoding	Handling missing data	Data Aggregation	Feature Engineering	Region of Interest (ROI) Extraction
Data source 1	Removed noisy OCR outputs, incorrect characters, and irrelevant text extracted from scoreboards Removed blurred, duplicate,umpire images.	Resized images to a fixed resolution for model input	Applied label encoding for class labels (Umpire, Non-Umpire) Used label encoding for event classes (FOUR, SIX, WICKET, NO ACTION).	Records with unreadable or missing player names were either corrected manually or removed.	Grouped detected events by player and over.	Extracted pose-related and visual features from umpire images.	Cropping scoreboard area Cropping umpire region only
	Outlier detection/ Handling	Date/ Time Transformation	Data Integration	Data Mapping	Data Aggregation	Class Balancing	Frame Sampling
	Removed false event detections	Converted frame numbers into time-based event windows. Converted extracted timestamps into a standard HH:MM:SS format.	Merged OCR player data and event detection results using timestamps.	Mapped OCR-extracted text to corresponding batsman and bowler labels.	Grouped detected events by player	Ensuring similar number of samples per class	Extracting frames at fixed intervals

3.2. Scalability

1. **Data Source 1**
   - **Player Identification dataset** - The dataset size is sufficient for player identification and player-event linking tasks. It is highly scalable, as additional matches can be processed to extract more OCR records with minimal manual effort.
   - **Umpire,non umpire dataset -**The dataset size is adequate for training and validating umpire detection models. Scalability is supported by capturing additional screenshots from new matches under different conditions.
   - **Umpire event detection dataset -**The dataset provides sufficient samples for multi-class event classification. It is scalable by adding more labeled images for additional events or increasing class balance

3.3. Feature extraction

Objective 1- IT22141606

<u>Key Features extracted</u>

1. **Player Identification from Scoreboard Text**

   - **OCR Text Features:** Cleaned textual content extracted from scoreboard regions using Optical Character Recognition (OCR).
   - **Token-Based Text Features:** Alphabetic tokens filtered by length, frequency, and stop-word removal to identify potential player names.
   - **Named Entity Features**: Player names extracted using a trained Named Entity Recognition (NER) model to distinguish batsmen and bowlers.

- **Frequency-Based Features:**
  Repeated name occurrences across consecutive frames used to improve reliability of detected players.

- **Temporal Features:**
  Frame timestamps and time continuity used to track player presence across overs.

These features were combined using an ensemble approach (NER + unsupervised token extraction + roster matching) to improve robustness.

2. **Umpire Detection**
   - **Spatial Image Features:**
     Cropped regions of interest (ROI) extracted from central and side areas of the frame to avoid scoreboard interference.

   - **Deep Visual Features:**
     High-level image features extracted using a convolutional neural network (EfficientNet-B0).

   - **Confidence Features:**
     Probability scores indicating the likelihood of an image containing an umpire.

   - **Multi-Crop Consensus Features:**
     Detection confidence aggregated across multiple crops to reduce false positives.

     Only frames exceeding a confidence threshold were selected for further event classification.

3. **Umpire Event Classification (4, 6, Wicket, No Action)**
   - **Pose Classification Features:**
     Visual pose representations extracted using a deep learning model (ResNet-18).

- **Gesture Confidence Scores:**
  Softmax probabilities used to measure confidence of each event class.

- **Margin-Based Features:**
  Difference between top two class probabilities used to reduce ambiguous predictions.

- **Event Priority Features:**
  Priority rules applied to resolve conflicts between overlapping event predictions.

- **Temporal Clustering Features:**
  Consecutive frame predictions grouped into clusters to stabilize event detection.

  Final events were generated using majority voting within clusters.

4. **Player–Event Linking and Timeline Construction**

- **Nearest Timestamp Features:**
  Events linked to the closest scoreboard frame based on time difference.

- **Temporal Voting Features:**
  Player predictions aggregated across short time windows to reduce misclassification.

- **Role-Based Features:**
  Separate tracking of batsman 1, batsman 2, and bowler roles.

- **Event Metadata Features:**
  Event type, confidence score, start time, and end time stored for each event.

These features enabled accurate player-wise event mapping and structured match timelines.

5. **Additional Features-Audio & Semantic Search**
   - **Audio Transcript Features:**
     Commentary text extracted using speech-to-text and segmented by time.
   - **Semantic Embedding Features:**
     Sentence-level embeddings generated for transcripts and events to support semantic search.
   - **Indexing Features:**
     Vector similarity indices created to allow fast retrieval of relevant match moments.

Feature Transformation

Feature extraction combined visual, textual, temporal, and semantic attributes to transform unstructured cricket videos into structured, searchable data. The extracted features directly support player-based video segmentation, accurate event detection, and efficient match navigation.

4. Modelling and Results

Objective 1- IT22141606

This section presents the key findings obtained from analyzing cricket match videos using computer vision, optical character recognition, and machine learning techniques. The results address the research objectives of player identification, umpire detection, and key event classification, based on systematic data exploration, model training, and integrated analysis pipelines applied throughout the study.

Modelling Approach

Multiple modelling techniques were employed to analyze different aspects of cricket match videos:

- **OCR-based text extraction models** were used to extract scoreboard information such as batsman and bowler names from video frames.

- **Named Entity Recognition (NER) and rule-based text processing** were applied to classify extracted text into batsman and bowler roles.

- **Convolutional Neural Network (CNN)**
  - **EfficientNet-B0 models** were used to detect the presence of umpires in video frames.
  - **ResNet-18 models** -were used to classify umpire gestures into FOUR, SIX, WICKET, and NO ACTION events.

- **Temporal clustering and timestamp-based aggregation techniques** were used to stabilize predictions and reduce false detections.

- Outputs from all models were integrated to generate **player-based video segments and event timelines**.

4.1. Key Insights:

1. OCR and Player Identification Dataset Analysis

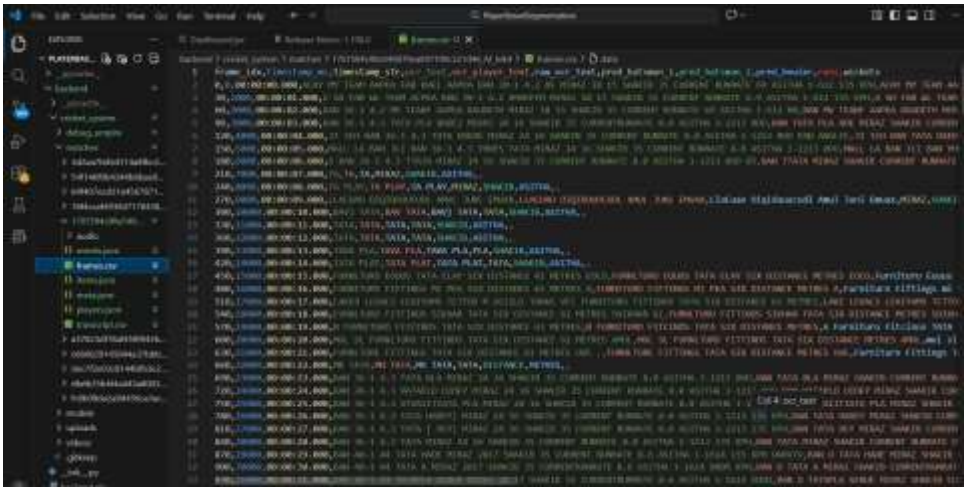
Exploratory analysis of the OCR dataset showed that scoreboard text frequently contained noise due to motion blur, low resolution, and varying broadcast formats. However, repeated appearance of player names across consecutive frames enabled reliable identification.

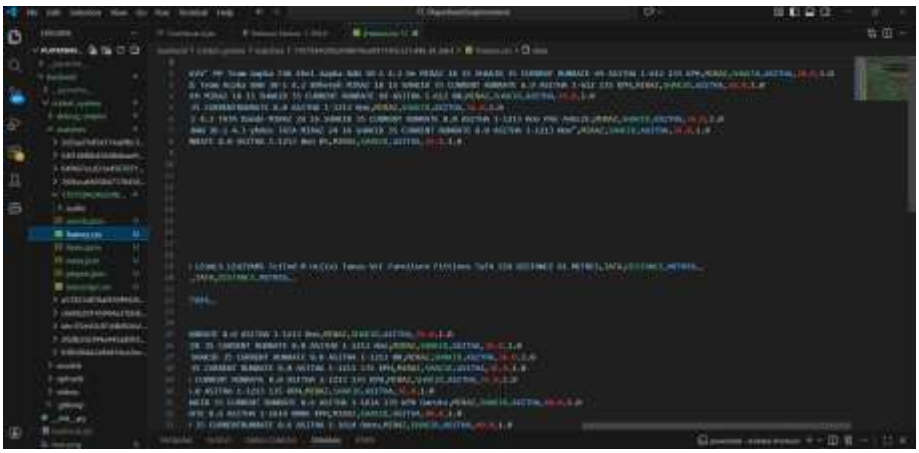
- Visualization of name frequency across time revealed clear patterns corresponding to striker changes and over transitions.

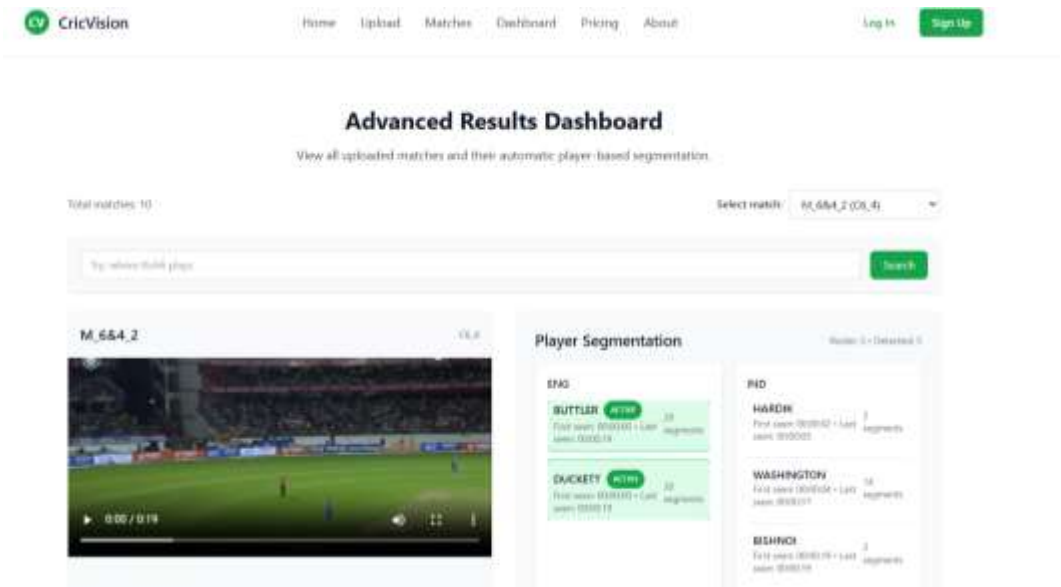
- Temporal consistency analysis demonstrated that aggregating predictions over short time windows significantly reduced incorrect player identification.

- Combining OCR with NER and rule-based validation improved overall accuracy compared to using OCR alone.

CSV file with predicted players







2. Umpire vs Non-Umpire Dataset Analysis

Analysis of the umpire detection dataset revealed balanced representation between umpire and non-umpire images.

- Visualization of classification confidence scores showed a clear separation between umpire and non-umpire classes.
- Detection performance improved when region-of-interest cropping was applied to remove irrelevant background information.
- Frames with centered and fully visible umpires produced higher detection confidence compared to partially occluded views.

3. Umpire Event Detection Dataset Analysis

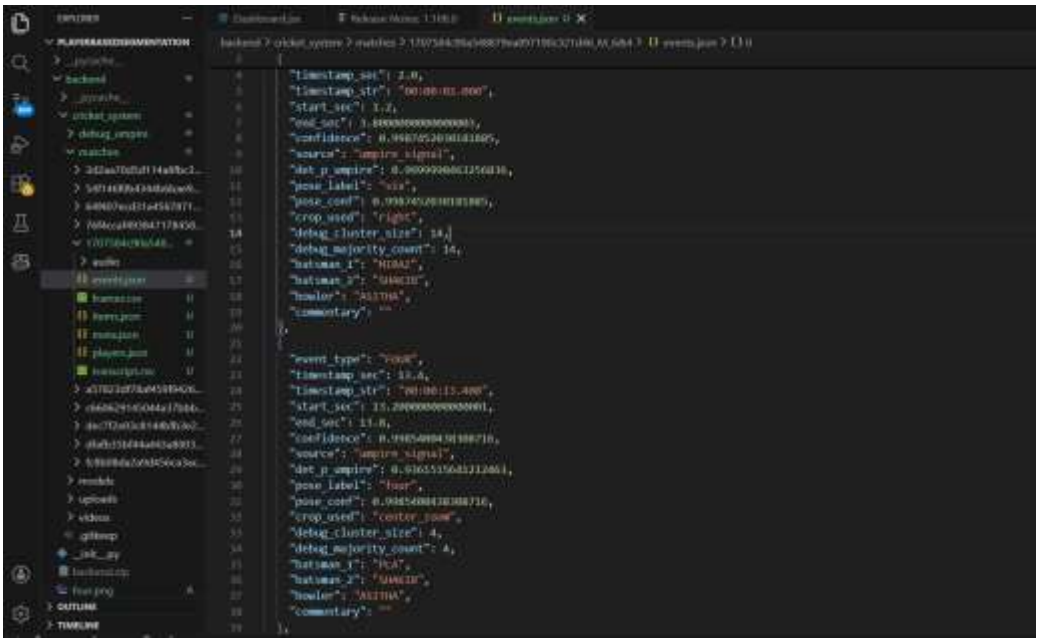
The event classification dataset exhibited class imbalance, with NO ACTION frames being more frequent than FOUR, SIX, and WICKET events.

- Class distribution plots highlighted the need for careful model tuning and balanced sampling.

- Gesture-based event classification achieved stable predictions when temporal clustering was applied.

- Confusion matrix analysis showed that FOUR and SIX gestures were occasionally misclassified due to similar arm positions, while WICKET signals were more distinct.

Detected umpire events



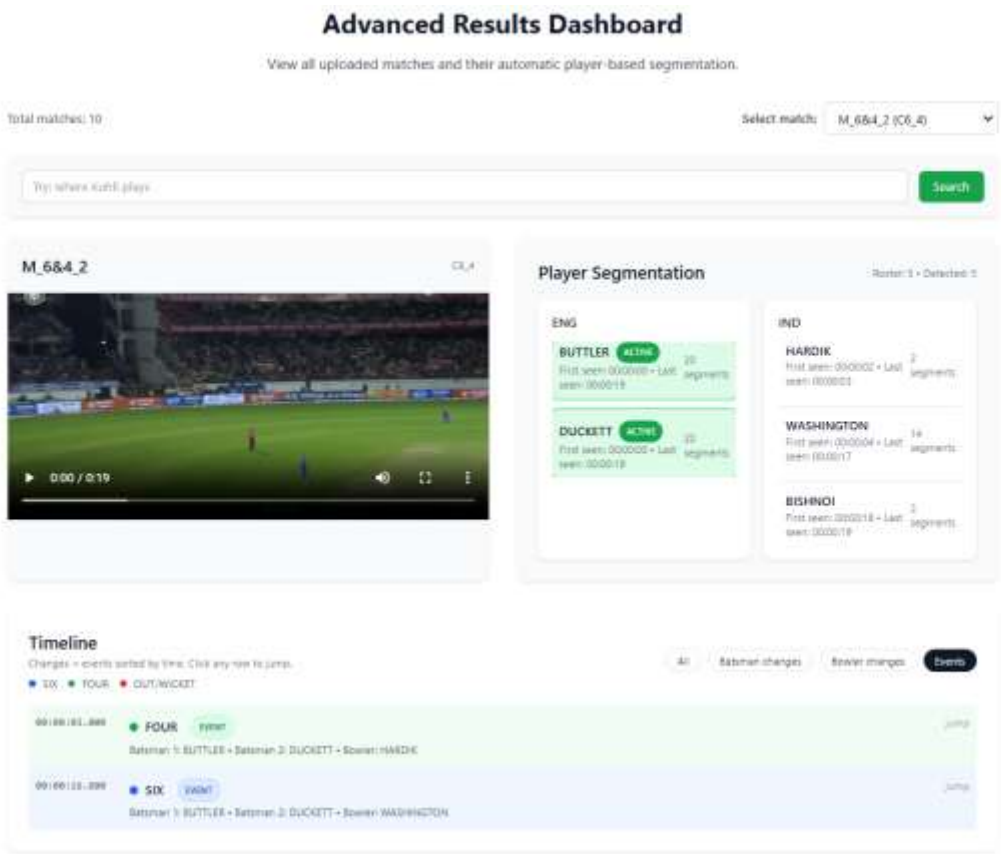
4. Player–Event Mapping and Timeline Construction

By integrating OCR-based player identification with event detection results:

- Detected events were successfully linked to active batsmen using nearest timestamp matching.

- Event timelines enabled direct navigation to player-specific highlights.

- Aggregated event windows reduced duplicate detections caused by camera replays and slow-motion shots.



4.2. Challenges Faced During Data Analysis:

- **Class Imbalance:** Certain event types, such as sixes or rare wicket types, were underrepresented in the umpire pose dataset. This required careful thresholding, multi-crop detection, and temporal clustering to prevent biased event detection.

- **Computational Constraints:** Processing full-length cricket videos, extracting frames, running OCR, deep learning-based umpire detection, pose classification, and speech transcription demanded significant computation and memory resources. Training and inference on multiple models required GPU acceleration for efficiency.

- **Scene Ambiguity:** In some video frames, umpires or players were partially occluded, poorly lit, or partially outside the frame, reducing the reliability of pose or player detection. Multi-crop strategies and confidence thresholds were applied to mitigate this.

- **Data Integration Complexity:** Combining multimodal features, including OCR-based player identification, umpire pose events, frame timestamps, and audio transcript segments, into a unified match timeline required careful temporal alignment, feature mapping, and smoothing to maintain accuracy across all data sources.

5. References

[1] J. Carreira and A. Zisserman, “Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4724–4733.

[2] H. Tang, V. Kwatra, M. E. Sargin, and U. Gargi, “Detecting highlights in sports videos: Cricket as a test case,” in *Proc. IEEE International Conference on Multimedia and Expo (ICME)*, 2011, pp. 1–6.

[3] J. Guo, C. Gurrin, S. Lao, C. Foley, and A. F. Smeaton, “Localization and recognition of the scoreboard in sports video based on SIFT point matching,” in *Proc. 17th International Conference on Advances in Multimedia Modeling (MMM)*, LNCS 6524, 2011, pp. 337–347.

[4] S. Nandyal and S. L. Kattimani, “Cricket event recognition and classification from umpire action gestures using convolutional neural network,” *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 13, no. 6, 2022.

[5] G. M. Park, H. I. Hyun, and H. Y. Kwon, “Multimodal learning model based on video–audio–chat feature fusion for detecting e-sports highlights,” *Applied Soft Computing*, vol. 126, 2022, Art. no. 109285.